iLearn Practice Exam

10BR · Chapter / Topic 7

Cambridge IGCSE Computer Science 0478

Paper 2 Algorithms, Programming and Logic

Chapter 7: Algorithm Design and Problem-solving

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: **75**

Marks obtained: ..... / 75

INSTRUCTIONS

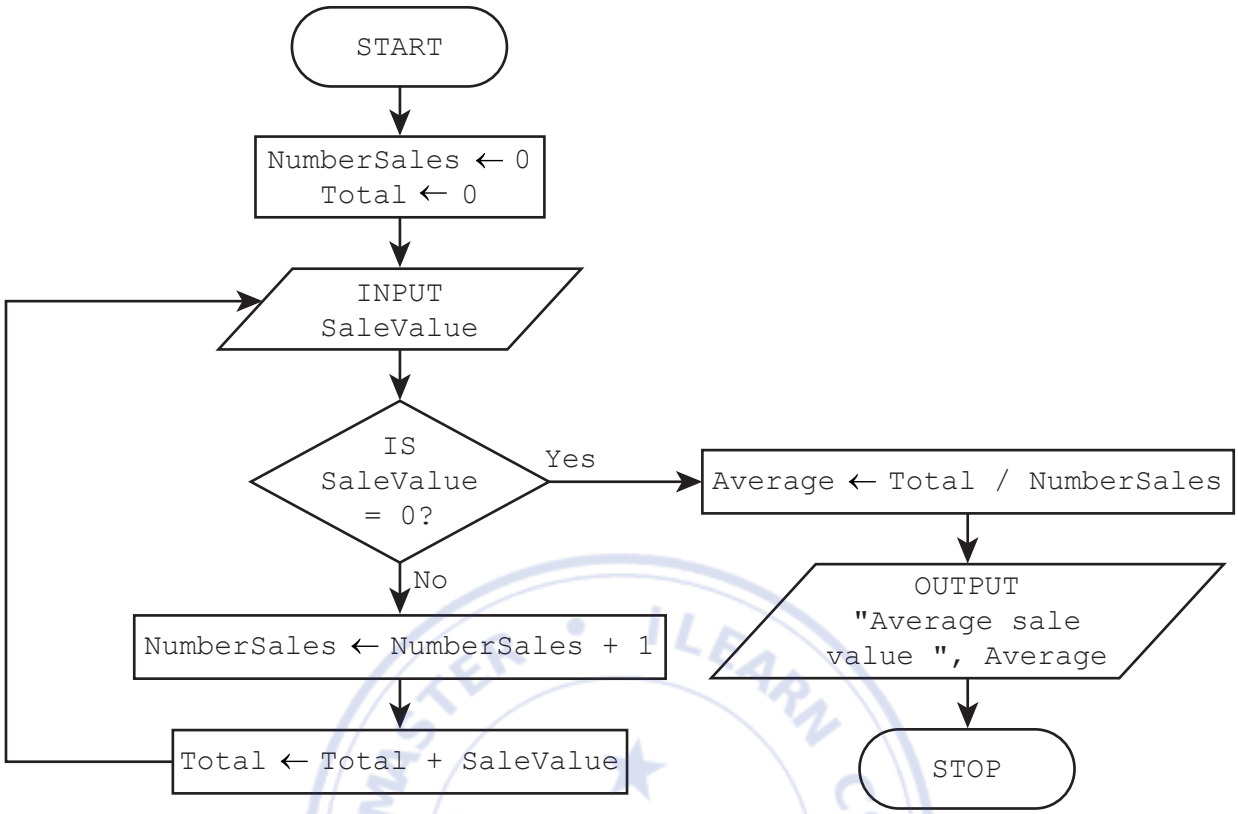
· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 75.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 10 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

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1 This flowchart represents an algorithm to find the average value of a number of sales.



(a) Complete the trace table using this data:
5.50, 3.40, 6.25, 3.85, –11.00, 0

NumberSales	Total	SaleValue	Average	OUTPUT

[4]

(b) Identify the error in the algorithm and describe how to correct it.

Error .....

.....

Correction .....

.....

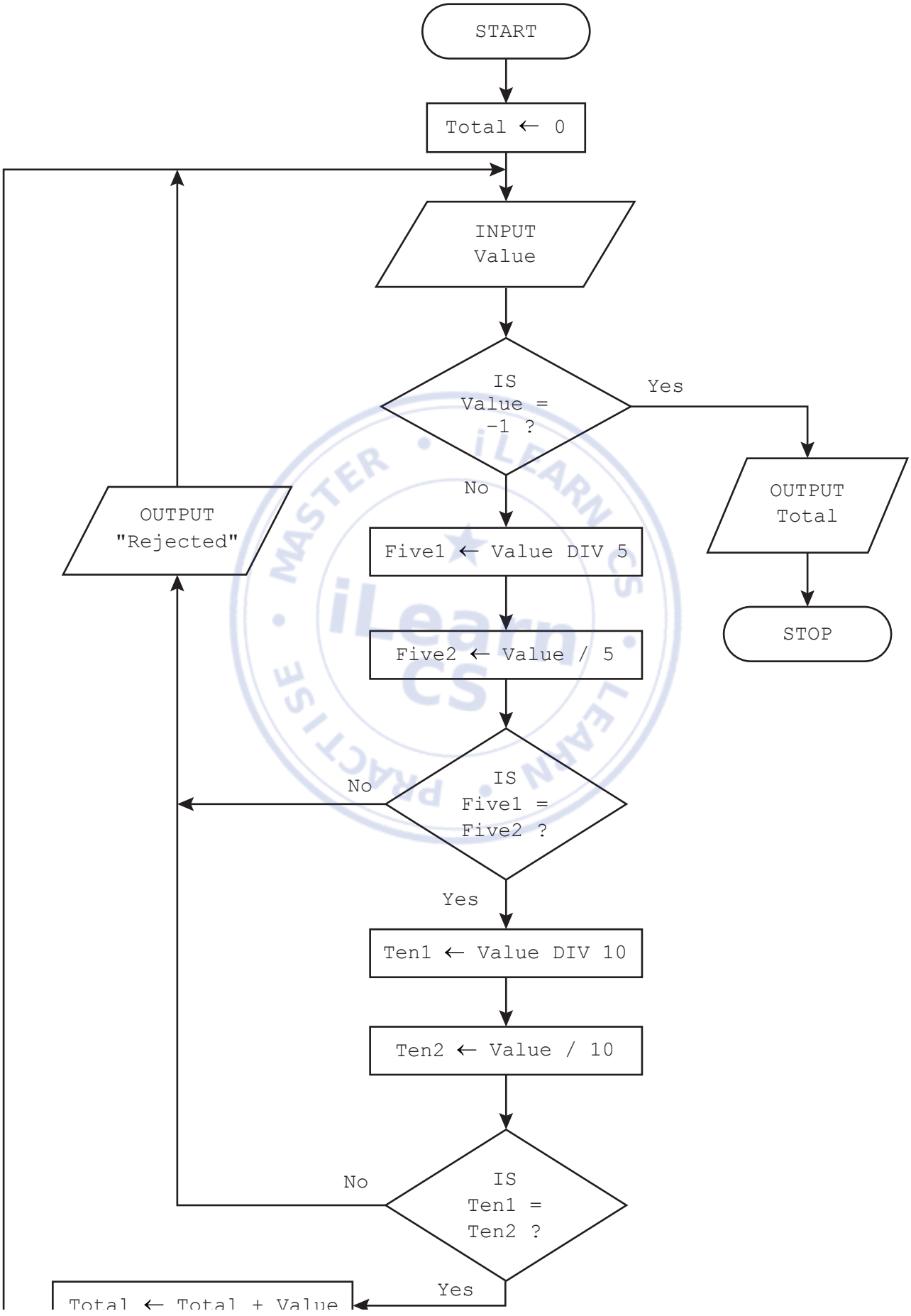
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[3]

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2 The flowchart represents an algorithm.
An input of –1 will terminate the algorithm.



(a) Complete the trace table for the input data:

5, 50, 52, 555, 57, 500, –1, 5500, 55

Total	Value	Five1	Five2	Ten1	Ten2	OUTPUT

[6]

(b) Describe the purpose of the algorithm.

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..... [2]

3 An algorithm has been written in pseudocode to check if a new password is in a list of previously used passwords `OldList[]`
If the password is **not** found, the new password will be stored at the end of the list to replace `"XXXX"` already stored there.

```
01 OUTPUT "Enter your new password "
02 INPUT NewPassword
03 Posn ← 1
04 Found ← FALSE
05 REPEAT
06     IF Password = OldList[Posn]
07       THEN
08         Found ← TRUE
09       ELSE Posn ← Posn + 1
10     ENDIF
11 UNTIL Found AND OldList[Posn] = "XXXX"
12 IF Found
13   THEN
14     OUTPUT "Password has been used before"
15   ELSE
16     INPUT "New password accepted"
17     OldList[Posn] ← NewPassword
18 ENDIF
```

(a) Identify the **three** errors in the pseudocode and suggest corrections.

Error 1 .....

Correction .....

.....

Error 2 .....

Correction .....

.....

Error 3 .....

Correction .....

.....

[3]

(b) Complete this flowchart for the corrected algorithm:





[6]

4 **(a)** **Four** test data types and **five** descriptions are shown.

Draw **one** line to link each test data type to its most appropriate description.

Test data type	Description
abnormal	a value that is accepted
boundary	a value that is the highest or lowest value to be accepted and the corresponding lowest or highest value to be rejected
extreme	a value that is the highest or lowest value to be rejected
normal	a value that is rejected
	a value that is the highest or lowest value to be accepted

[4]

(b) An algorithm has been written to test if an integer that is input is in the range 5 to 10 inclusive. Identify an example of suitable test data for each test data type.

Abnormal .....

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Boundary .....

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Extreme .....

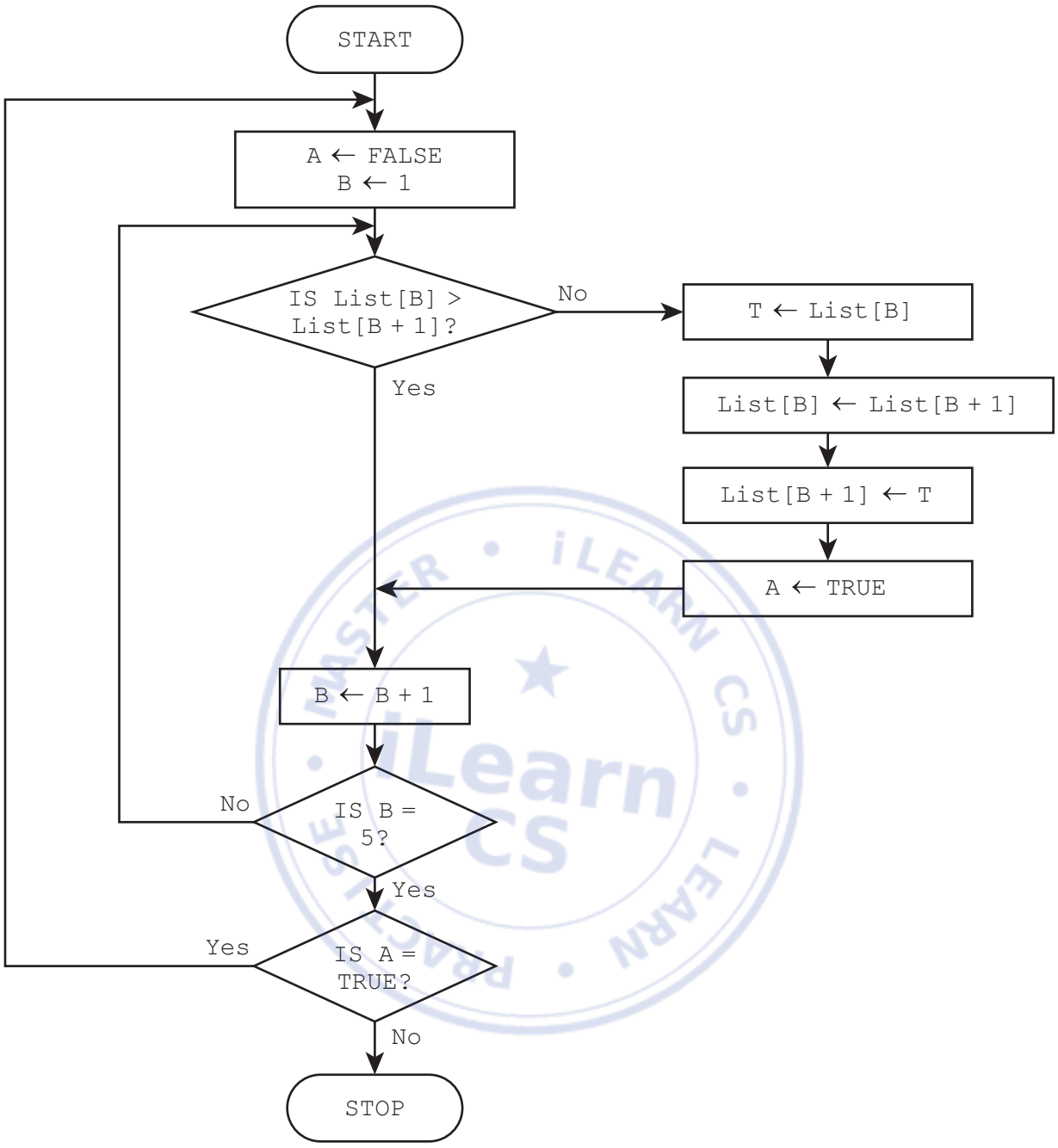
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Normal .....

.....

[4]

5 This flowchart represents an algorithm.



(a) The array `List[1:5]` used in the flowchart contains this data:

List[1]	List[2]	List[3]	List[4]	List[5]
15	17	20	5	9

Complete the trace table using the data given in the array.

A	B	List[1]	List[2]	List[3]	List[4]	List[5]	T
		15	17	20	5	9	

[5]

(b) Describe what the algorithm represented by the flowchart is doing.

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6 One stage of the program development life cycle is the analysis stage.

Identify and describe **two** other stages of the program development life cycle.

Stage .....

Description .....

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Stage .....

Description .....

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[6]

7 **(a)** A presence check is a type of validation check.

State the purpose of a presence check.

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..... [1]

(b) A value input into a computer system must be an integer.

(i) Identify the validation check used to test whether the value input meets this requirement.

..... [1]

(ii) Write the pseudocode for an algorithm to allow a number to be input into an appropriate variable and check whether the number is an integer. If the number is an integer, it is accepted. If the number is **not** an integer, an error message is output. Re-input of the number is required until a valid number is input.

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..... [5]

8 Tick (✓) **one** box to complete the description of abstraction.

Abstraction is part of the analysis stage of the development life cycle. Abstraction means

A testing the solution to the problem. ☐

B writing a program using program code. ☐

C defining the sub-systems that make up the problem. ☐

D removing details from the problem that are not relevant. ☐

[1]

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9 A collector needs a video library system for a collection of videos stored on 4K Blu-ray discs, standard Blu-ray discs, DVD, or as digital downloads.

The two-dimensional (2D) array `Video[]` is used to store, for each video item, the title, the format of the video (4K Blu-ray disc, standard Blu-ray disc, DVD, or digital download), the year it was released stored as a string and a storage code to represent where the video can be found in the collection, for example:

```
Video[1,1] is "Macbeth"
Video[1,2] is "digital download"
Video[1,3] is "2015"
Video[1,4] is "DG276"
```

The two-dimensional (2D) array `Results[]` is used to store the results of a search to find a specific video. The search uses the video title, and if the video is found, its data is copied from the array `Video[]` to the array `Results[]`. The search continues and if other videos with the same title are found, this data is also copied to the array `Results[]`. The search ends when the end of the data is reached.

Write a program that meets the following requirements:

- The video library array needs to be initialised with the null string (""), allowing for up to 10 000 records. The array `Results[]` must be initialised before every search with the null string and must be able to hold up to 20 search results with the same title.
- Create a menu to add a new video to the library to search for an existing video by title, or to stop, with validation of the input.
- When the user enters a new video, the following data is stored in the first available location of the relevant array:
  - video title
  - format (4K Blu-ray disc, standard Blu-ray disc, DVD, digital download)
  - year of release as string
  - storage code.
- Allow input of data for another video if required.
- When searching for an existing video, if a match is found, transfer all the data for that video into the array `Results[]`, and continue to search until all videos with the same title have been found.
- Output the results from the array `Results[]`
- If the video title is not found, output a suitable message.
- The system returns to the menu after completing the input or the output, until the user chooses to stop.

You must use pseudocode or program code and add comments to explain how your code works.

You do **not** need to declare any arrays, variables or constants. You may assume that this has already been done.

All inputs and outputs must contain suitable messages.

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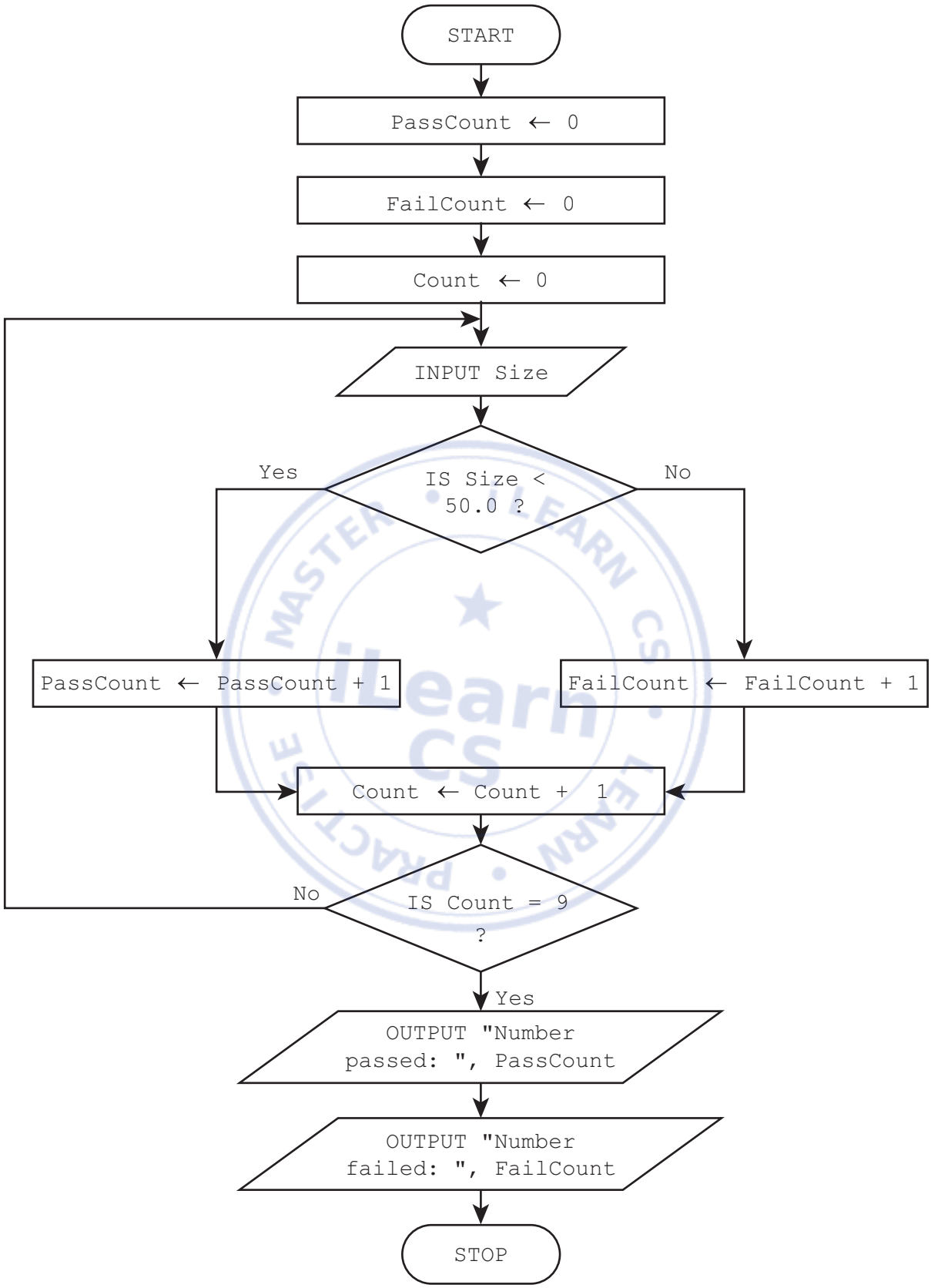
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[15]

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10 This flowchart represents an algorithm.



(a) Complete the trace table for the algorithm, using the input data:

49.8, 48.7, 62.5, 49.9, 30.0, 58.2, –12.5, 50.0, 26.7, –49.2, 58.9

PassCount	FailCount	Count	Size	OUTPUT

[5]

(b) Describe how you would change the flowchart to prevent negative numbers from being input.

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..... [2]

END OF QUESTION PAPER

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